

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Benefits and Risks of Conversion of Existing Adolescent Kidney Transplant Recipients Aged 12 to <18 Years to a Belatacept-based Immunosuppressive Regimen as Compared to Continuation of a Calcineurin Inhibitor-based Regimen, and Their Adherence to Immunosuppressive Medications **Official Title:** A Prospective, Open-label, Multicenter, Randomized Study to Evaluate the Benefits and Risks of Conversion of Existing Adolescent Renal Allograft Recipients Aged 12 to Less Than 18 Years of Age to a Belatacept-based Immunosuppressive Regimen as Compared to Continuation of a Calcineurin Inhibitor-based Regimen, and Their Adherence to Immunosuppressive Medications **Short Title/Acronym:** Conversion to Belatacept-based Immunosuppression in Adolescent Renal Transplant Recipients **Protocol Number:** IM103-402 **NCT Number:** NCT04877288 **Sponsor/Company:** Bristol Myers Squibb **Investigational Product:** Belatacept (Nulojix) **Phase of Development:** PHASE3 **Medical Monitor:** [Bristol-Myers Squibb Clinical Operations Lead]

2. Study Synopsis & Rationale

2.1 Study Objective / Summary Purpose/Primary Objective: The purpose of this study is to evaluate the benefits and risks of conversion of existing adolescent kidney allograft recipients aged 12 to less than 18 years of age to a belatacept-based immunosuppressive regimen as compared to continuation of a calcineurin inhibitor-based regimen and their adherence to immunosuppressive medications.

2.2 Background & Rationale Kidney transplantation requires the use of anti-rejection medications for transplant survival. Calcineurin inhibitors (CNIs) have varied safety profiles and off-target effects (nephrotoxicity, cardiovascular, or metabolic toxicities) that may adversely impact the quality of life and result in non-compliance. Adolescent kidney transplant recipients are at a higher risk of graft rejection due to poor medication compliance. Belatacept offers a targeted immunosuppressive alternative that can avoid CNI-related toxicities while offering similar short-term patient and graft survival, with administration via intravenous infusion every 4 weeks potentially improving adherence.

3. Study Design & Methodology

3.1 Design Framework Study Type: INTERVENTIONAL **Phase:** PHASE3
Status Name: RECRUITING **Status Description:** Currently recruiting participants **Control Type:** Active Comparator (Continuation of CNI-based regimen) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 102 evaluable adolescent patients **Number of Sites:** Multicenter (Global, including US and Europe) **Estimated Study Start:** 21-Jul-2021 **Estimated Primary Completion:** 29-Jun-2034

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female adolescents 12 to less than 18 years of age. 2. Recipients of a renal allograft from a living or deceased donor transplanted at least 6 calendar months prior to enrollment. 3. Receiving a stable regimen of a calcineurin inhibitor (CNI), with mycophenolate mofetil (MMF) or enteric-coated mycophenolate sodium/mycophenolate mofetil (EC-MPS/MPA), with or without daily corticosteroids for ≥ 30 days prior to randomization. 4. Clinically stable renal function during the 12-week period prior to screening, in the opinion of the investigator and based on protocol-defined criteria for proteinuria and estimated glomerular filtration rate (eGFR). 5. Serologic evidence of past exposure to Epstein-Barr virus (EBV) and current absence of EBV DNA replication at or prior to renal transplantation and during the Screening period. 6. Completion of an initial course of SARS-CoV-2 vaccination per local standard of care, a minimum of 6 weeks prior to enrollment.

4.2 Exclusion Criteria 1. Recipients with EBV serostatus negative or unknown at screening or at transplant. 2. Treatment for biopsy-proven acute rejection (BPAR) of any degree of severity within 6 calendar months prior to enrollment. 3. Biopsy-confirmed antibody-mediated acute rejection at any time with the current allograft. 4. Banff 97 grade IIA or higher acute cellular rejection (or equivalent), or treatment with plasmapheresis or rituximab for any acute rejection at any time with the current allograft. 5. Current evidence or past history of active or inadequately treated latent tuberculosis (TB) infection. 6. Previously treated with belatacept or previously enrolled in a belatacept trial with their present allograft. (*Note: Other inclusion/exclusion criteria apply*)

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Arm 1 (Investigational): Belatacept administered on Days 1, 15, 29, 43, 57, and then every 28 days. The CNI dose is tapered and stopped during the 12-24 weeks after Day 1. Arm 2 (Control): Continue CNI

treatment. **Route of Administration:** Intravenous (IV) infusion for belatacept; Oral for standard CNIs and background immunosuppressants. **Duration of Treatment:** Long-term follow-up through to study completion (Approx. 13 years total duration).

5.2 Concomitant & Prohibited Therapy Permitted: Mycophenolate Mofetil (MMF), Enteric-coated mycophenolate sodium (EC-MPS), and corticosteroids. **Prohibited:** Concurrent investigational therapies and live vaccines that conflict with standard post-transplant immunosuppression protocols.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Prior to Day 0	Informed Consent, EBV serostatus check, Medical History, eGFR Baseline Labs
Baseline	Day 1	Randomization, First Belatacept Dose (Arm 1) or CNI maintenance, Quality Audit
Interim	Days 15, 29, 43, 57, then Q28 Days	Belatacept infusion (Arm 1), CNI taper management (Weeks 12-24), Efficacy/Safety Labs
End of Study	Year 2034	Final Vital Signs, Exit Interview, IP Accountability, Final Graft/Survival Check

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Proportion of participants who survive with a functional graft with estimated glomerular filtration rate (eGFR) > 30 mL/min/1.73 m² (updated Schwartz formula) at 24 months post-randomization.

7.2 Secondary Outcomes 1. Participant and graft survival: Proportion of participants who survive with a functioning graft. 2. Participant and graft survival: Proportion of participants who survive. 3. Participant and graft survival: Proportion of participants who experience death-censored graft loss. *(Additional exploratory parameters evaluate belatacept whole blood trough concentrations, metabolic parameters, hypertension management, de novo anti-HLA donor-specific antibody formation, and Tanner staging/linear growth).*

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient interviews, physical examinations, and laboratory assessments at each 28-day infusion visit. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness; close monitoring for post-transplant lymphoproliferative disorder (PTLD), infections, and acute rejection.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE/SAE evaluations. **Sample Size Justification:** \$N=102\$ subjects powered to evaluate the non-inferiority or superiority of the conversion regimen compared to standard of care in a rare pediatric/adolescent transplant subpopulation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits focusing on proper CNI tapering and adverse event recording.

11. Study Identifiers & Database Metadata

Primary ID (NCT Number): NCT04877288 **Posting ID:** 458611262807086306 **Phase ID:** PHASE3 **Criteria Type:** PHASE_REQUIREMENT (Criteria Text: Trial must be in PHASE3) **Source Level:** 5

11.1 Associated Drug Mappings (Database Extractions) (*Note: Included for internal database mapping completeness*) * **Drug Name / Drug Preferred Name:** phenol * **ATC Code:** N01BX03 * **Additional ATC Codes:** R02AA19, C05BB05, D08AE03 * **RxNorm Code:** N/A (Blank) * **Trade Names:** Dermagesic Itch Relief, Chloraseptic Max Spray, Stingeze, Chigarid, Phenol EZ, Ora Relief, Derma Cas, Cheracol Sore Throat, Chloraseptic Spray, Castellani Paint, Campho-Phenique, Cepastat, Mycinette Sore Throat Spray, Pain-A-Lay Gargle, Unguentine, Ulcerease, Clearsol, Hycolin, ChapIce with Phenol, StingEze w/ Camphor, Off! Explore, Stingeze Max, Unguentine Maximum Strength

12. Clinical Framework (Transplant Specific & Ontologies)

Disease Areas / Conditions: Renal Allograft Recipients **Name:** Renal Allograft Recipients **Preferred Name:** Implantation of renal allograft with recipient nephrectomy **Preferred UMLS Name:** Immunosuppressive regimen **Semantic Type:** Therapeutic or Preventive Procedure **Definition:** Renal allograft recipients are individuals who have undergone a kidney transplant, meaning they have received a kidney from another human donor. This procedure is performed to replace failing native kidneys, typically due to end-stage renal disease, restoring vital kidney functions such as filtering waste products from the blood, maintaining fluid and electrolyte balance, and producing hormones. These recipients require lifelong immunosuppressive medication to prevent their

immune system from rejecting the transplanted organ. **MeSH Code:**
D016922 **SNOMED ID:** 712061002 **SNOMED Hierarchy:** 105658007 |
Recipient of transplant | Recipient of transplant (finding)

13. Study Procedures & Methodology

Management Pathway: Belatacept conversion following a 3-6 month CNI taper. **Optimization Target:** Avoidance of CNI-related nephrotoxicity and improvement of patient adherence. **Education Protocol (Patient):** Counseling on medication adherence and infusion schedules.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 13 Years (Until June 2034) **Onsite Visit Cadence:** Days 1, 15, 29, 43, 57, then every 28 days.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					